

## Customer Churn Prediction Using Machine Learning

**Project Overview** Customer churn prediction helps businesses identify customers likely to leave. This project uses machine learning to predict churn probability.

**Problem Statement** Customer churn causes revenue loss. Early prediction helps retention strategies.

**Technologies Used** Python, Pandas, NumPy, Scikit-learn, Jupyter Notebook

**Dataset Description** Customer demographics, service usage, billing details. Target: Churn (Yes/No)

**Model** Random Forest Classifier

**Sample Result** Prediction: No Churn Probability of Churn: 0.22

**Conclusion** Machine learning enables accurate churn prediction and business decision support.